

[Logo: Hackles]

Concepts Lab

Trust your hackles. Pause · Check · Tell.

FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-in-inked, in the order the lessons take them.

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OCTOBER 2026

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Hackles • Cybersecurity Concepts Lab for Teachers

Demo 1: Why Long Passphrases Win (The Dice Crack-Timer)

Trust your hackles. Pause • Check • Tell

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE

Big idea	Every extra random piece you add to a secret multiplies the guessing work. Length and randomness beat cleverness.
Grade bands	K-2 (the "secret handshake" idea), 3-5 (full demo), 6-8 (full demo + math), 9-12 (full demo + bits and attack models)
CSTA-style concepts	Networks and the Internet (cybersecurity): passwords and authentication, tradeoffs of security measures (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute version and 45-minute version
Cost and tech	Free to a few dollars (dice). No devices. A calculator helps for the 6-12 level-up.
Teachers do it first	Yes. Do Part A (about 20 minutes) before you teach.

MAKE IT STICK

Try This Now (60 seconds, no prep): Hide one die roll under a cup. Your neighbor guesses (at most 6 tries). Now hide two dice (at most 36). Ask: "What happens at ten dice?" You just felt a search space grow.

01 The big idea

A **PASSWORD IS A** lock whose strength is "how many wrong guesses would someone have to try before finding the right one?" That number is the **search space**. Computers guess extremely fast, so the only reliable defense is a search space too big to finish in a lifetime.

The surprise for most adults: the biggest search spaces come from **length plus true randomness**, not weird symbols. Four or five random words beat "P@ssw0rd!" by an astronomical margin, and they are easier to remember.

02 What a teacher should know (plain language)

2.1 TWO VERY DIFFERENT KINDS OF GUESSING

- **Online guessing.** Someone types guesses into a real login page. The website can slow them down, lock the account, or notice. A few thousand guesses per day might be all they can make. Here even a mediocre password often survives, which is why "lockout after 10 tries" is such a useful defense.
- **Offline guessing.** After a company is breached, attackers may hold a stolen file of *scrambled* passwords (Demo 2). They can then guess on their own computers with no lockout, at billions of guesses per second against fast scrambling methods. Long random passphrases are built for this case. [Teaching assumption, not a measured fact: real speeds depend on hardware and on the hashing method. The 10 billion per second used below is a round, deliberately plain figure; no primary benchmark was checked in Round 3B]

TEACH BOTH. MANY "MYTHS" about passwords come from mixing them up.

2.2 SEARCH SPACE, IN ONE LINE

If each position in a secret can be one of **N** options and the secret has **L** positions, the number of possible secrets is **N to the power L**. On average a guesser finds it after trying about **half** of them.

SECRET	OPTIONS PER POSITION	POSITIONS	POSSIBLE SECRETS
4-digit PIN	10	4	10,000
6 random lowercase letters	26	6	about 309 million
8 random lowercase letters	26	8	about 209 billion
3 random words from a 7,776-word list	7,776	3	about 470 billion
4 random words	7,776	4	about 3.7 quadrillion

SECRET	OPTIONS PER POSITION	POSITIONS	POSSIBLE SECRETS
8 random characters from a full keyboard set (94 symbols)	94	8	about 6.1 quadrillion
5 random words	7,776	5	about 28 quintillion (2.8×10^{19})
6 random words	7,776	6	about 2.2×10^{23}
12 random characters from 94 symbols	94	12	about 4.8×10^{23}

The 7,776-word list is the Diceware idea: five dice rolls pick one word ($6 \times 6 \times 6 \times 6 \times 6 = 7,776$). The Electronic Frontier Foundation publishes improved word lists in this format: a 7,776-word long list and two 1,296-word short lists [VERIFIED 2026-10-05: eff.org/deeplinks/2016/07/new-wordlists-random-passphrases].

2.3 BITS (FOR GRADES 9-12 AND CURIOUS ADULTS)

"Bits of entropy" is just the number of times you would double a tiny search space: $\text{bits} = \log_2(\text{possible secrets})$. One random word from 7,776 is about 12.9 bits. Six words is about 77.5 bits. Eight random keyboard characters is about 52 bits. Each added bit doubles the work. **This only counts if the choices are truly random.** A phrase you invented has far fewer bits than its length suggests, because humans pick predictable words.

2.4 HONEST CRACK-TIME TABLE (OFFLINE, FAST HASH)

Assumption: an attacker makes **10 billion guesses per second** and finds the answer on average after trying half the space (the worst case takes twice as long). Real attackers may be faster or slower. A properly stored password (slow, salted hashing, Demo 2) can make each guess thousands to millions of times more expensive [the factor depends on the algorithm and settings; OWASP's current minimums are Argon2id with 19 MiB memory and 2 iterations, bcrypt work factor 10 or more, and PBKDF2-HMAC-SHA-256 with 600,000 or more iterations; VERIFIED 2026-10-05: cheatsheetseries.owasp.org].

SECRET	AVERAGE TIME AT 10 BILLION GUESSES PER SECOND
4-digit PIN	instant (a fraction of a millisecond)
8 random lowercase letters	about 10 seconds
3 random words (7,776 list)	about 24 seconds
4 random words	about 2 days
8 random characters from 94 symbols	about 3.5 days
5 random words	about 45 years
6 random words	about 350,000 years
12 random characters from 94 symbols	about 750,000 years

I

Read this table honestly: **3 words is weak, 4 is borderline for important accounts, 5 to 6 is strong for a password you must memorize.** The headline is not "words good, symbols bad." It is "random and long".

2.5 WHAT ACTUALLY CAUSES MOST REAL ACCOUNT BREAK-INS

Most stolen accounts are not cracked at all. They come from **reuse** (a leaked password tried at other sites, called credential stuffing), **phishing** (Demo 3) and **malware** (Demo 11). A strong, unique password defends against cracking and reuse. A password manager (a locked app that remembers long random passwords) is the practical answer for most accounts; a memorized passphrase is for the few secrets you must keep in your head: the manager's master password, the device login, the email account.

2.6 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"Replace a with @ and o with 0 and I'm safe."	Cracking tools try these swaps first. They add almost no strength. Randomness and length do.
"Change it every 30 days."	Forced frequent changes lead to weak patterns (Summer2025!, Summer2026!). Current US guidance (NIST SP 800-63B-4, August 2025) says services must not force periodic changes without evidence of compromise, and must check new passwords against known-breached lists [VERIFIED 2026-10-05: pages.nist.gov/800-63-4/sp800-63b.html]. Change it when there is a reason: a breach notice, a shared device, a lost phone.
"My famous quote or song lyric is long, so it's strong."	Quotes are in attackers' lists. The words must be chosen by dice, not by you.
"Nobody would bother attacking me."	Automated attacks do not choose people; they try everyone.
"Writing passwords down is always bad."	A written paper kept at home can be better than reuse. The threat is distance: a stranger online cannot read your desk drawer. Use judgment; do not stick it on the monitor at school.

03 Materials (zero to low cost)

- 10 to 30 six-sided dice (or numbered paper slips in a cup; one die per student or two per pair), index cards, a timer, chart paper.
- The **Mini Word List** below (one per pair), the student handout, and a calculator for grades 6-12 (optional).

Mini Word List (36 words, below). Print one per pair.

- A wall clock or timer. Chart paper.
- Calculator (optional, grades 6-12).
- The student handout (end of this file).

Mini Word List (36 words; roll two dice, first die = row, second die = column)

ROLL	1	2	3	4	5	6
1	apple	brick	cloud	dance	eagle	flame
2	giant	honey	igloo	jelly	kite	lemon
3	moose	noodle	otter	pillow	quilt	river
4	spoon	tiger	umbrella	violin	whale	yogurt
5	zebra	anchor	button	candle	drum	engine
6	feather	garden	hammer	island	jacket	ladder

Teacher note: 36 words is a toy list. The real Diceware list has 7,776 words. We use the small list so the whole class can see the idea on a single page. Say that out loud.

04 Part A: Teachers do it first (about 20 minutes)

You are the "crackers" and the "vault keepers".

- 1. Warm-up (3 min).** Write an invented password that looks strong and pass it face down to a colleague. It must not be one you use.
- 2. Dice Vault (7 min).** One person rolls 3 dice under a cup (216 possibilities); the others guess in order and count. Repeat with 4 dice (1,296) and with 10 dice (60,466,176: at one guess per second, almost two years to try them all).
- 3. Build a passphrase (4 min).** Roll two dice per word and build 4 words from the Mini Word List ($36 \times 36 \times 36 \times 36 = 1,679,616$ possibilities). Then swap in the real 7,776-word list: 3.7 quadrillion.
- 4. Cracker clock (6 min).** Fill in the worksheet below with a calculator.

Debrief: where did your intuition fail? Most people expect the 8-character mixed password to be far stronger than it is, and are surprised that 5 random words beat it by thousands of times.

05 Part B: Teaching it

THE CRACKER CLOCK WORKSHEET (TEACHER COPY)

SECRET	POSSIBILITIES	HALF OF THEM (AVERAGE)	TIME AT 1 GUESS PER SECOND	TIME AT 10 BILLION PER SECOND
4 dice	1,296	648	about 11 minutes	instant
8 dice	1,679,616	839,808	about 10 days	instant
12 dice	2,176,782,336	1,088,391,168	about 34 years	0.1 seconds
16 dice	2.8 trillion	1.4 trillion	about 45,000 years	about 140 seconds

(Rolling dice as a secret is a neat model: each die adds a factor of 6.

"Doubling the length squares the pain.")

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook: the teacher reads the script. Pairs get dice.
2-6	Round 1: each pair makes a 3-dice secret (hidden), partners guess in order and count. Then 4 dice (teacher judges "who finished first").
6-10	The Big Jump: the teacher rolls 10 dice secretly. Whole class tries for 90 seconds; show 60,466,176 and "almost two years to try them all at one guess per second".
10-13	Class chant and table: each extra die multiplies by 6. Passphrase mini-build: 3 words from the Mini Word List.
13-15	Exit slip: "A good passphrase is ____ and ____". Answer: long and random.

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-5	Hook and "write a password you think is strong" warm-up (fictional passwords only).
5-15	Dice Vault rounds (3, 4, 6 dice) with team scoring.
15-25	Passphrase Lab: pairs roll a 4-word passphrase using the Mini Word List; compute search space; swap with another pair to cross-check the math.
25-35	Cracker Clock: compute the table; discuss online vs offline; show the 10 billion per second thought experiment (below).
35-42	"Mystery Fix-It": the teacher shows 6 fictional password habits; teams decide fix-it advice (reuse, rotate, symbols, manager).
42-45	Exit ticket and handout.

06 Teacher script

Hook (read aloud, 2 minutes). "Imagine a lock on a treasure chest that has a tiny dial with the numbers 1 to 6. If I told you the code was one number, how many tries would it take you? Right, six at most. Two numbers? Thirty-six. Today we find out what happens when the lock gets longer, and why a computer that tries billions of guesses a second still gets stopped by a long, random secret. We will use dice, and by the end you will be able to explain it to someone at home."

The 10 billion thought experiment (45-minute version). "A fast computer can try 10 billion guesses a second. Our class makes about one guess every two seconds per student. To match that computer we would need about 670 million classrooms of 30 students. So we need secrets long enough that even billions of guesses a second do not finish. And remember: this only matters if someone can guess offline, away from the website's lockouts."

Close. "A strong password is long and random. A passphrase is a long, random password made of words. A password manager is a helper that remembers long random passwords so you don't have to. If someone ever learns your password, it's never your fault. Change it, tell an adult, and we fix it together."

07 Kid-friendly analogies by band

BAND	ANALOGY AND ACTIVITY
K-2	The secret handshake. A handshake with two moves is easy to copy; one with eight silly moves is very hard. "Long and silly beats short and simple." Use four animal pictures drawn from a bag as the "secret animal parade". Never ask for a real password; pretend ones only.
3-5	The dice vault and the combination lock. Each added dice number multiplies the possibilities by six. Build "silly sentence passphrases" from the Mini Word List. Compare a bike lock with 3 wheels (1,000) to one with 6 wheels (1,000,000).
6-8	The haystack. Each extra random word makes the haystack 7,776 times bigger. Work out the numbers with exponents. Discuss online vs offline attacks and why password reuse is the real danger.
9-12	Entropy and attacker models. Bits, log2, online throttling vs offline cracking, salts and slow hashes (link to Demo 2), password managers and passkeys (link to Demo 6). Evaluate NIST guidance.

08 Discussion questions

1. Which grew faster: adding one more die, or switching from letters to letters-and-symbols? Why?
2. Why is a famous quote a bad passphrase even though it is long?
3. Why can a website protect a short password from online guessing but not from offline guessing?

4. (9-12) Why does adding a die add a constant number of bits, no matter how many dice you already have?

09 Level-up challenge

- **The honest calculator.** Students build a spreadsheet: choose alphabet size and length, output search space and crack time at three guess rates (1,000 per second online-ish, 10 million per second slow hash, 10 billion per second fast hash) and explain which rate fits which attack. The rates are illustrative.
- **Pattern buster (9-12).** Explain, without writing any cracking tool, why a human-chosen password like "Summer2026!" has far fewer bits than its length suggests (season word, year, symbol).

10 Safety and ethics notes (defense only)

- Use only **invented** passwords in class. Never ask students to say, write or share a real one. If a student blurts one, say calmly "thank you for noticing; change it tonight."
- Do not teach or demonstrate password-cracking tools, and never run guesses against school logins, classmates' accounts or any website. Attacking accounts without permission is illegal and harmful (Demo 14). The dice game models guessing, not software tools.
- If a student mentions that someone knows their password or is using their account, treat it as a possible safety concern and follow your school's reporting process (see the box in the Index). Never shame a weak password: "You can upgrade it tonight."

11 Assessment idea

- **Exit ticket (all bands).** "Draw or write: Why is a 5-word dice passphrase stronger than a 9-character password with symbols? Use the words 'multiply' and 'random'."
- **Rubric.** 3 = states that each extra element multiplies possibilities and that randomness matters; 2 = states only that longer is better; 1 = states a myth (swap letters for symbols).

12 Sources and [VERIFY] list

- Diceware (Arnold Reinhold, 1995; date not checked) and EFF wordlists (the long list has 7,776 words and the short lists 1,296 each) [VERIFIED 2026-10-05: eff.org].
- NIST SP 800-63B-4 Digital Identity Guidelines, final August 2025 (password length, breached-password checks, no forced rotation) [VERIFIED 2026-10-05].
- Guess rates for modern GPUs against fast hashes (order of magnitude billions per second) [teaching assumption; not verified against a primary benchmark].
- Credential stuffing (trying leaked passwords on other sites) as a common route to account takeover [general security knowledge; no percentage is claimed, and the Verizon DBIR was not checked in Round 3B].

Student Handout: The Dice Crack-Timer

Name: _____ (no real passwords on this page, ever)

The big idea

EVERY EXTRA RANDOM PIECE multiplies the guessing work. Longer and random wins.

Try it

ROLL A DIE. A 1-dice secret has 6 possibilities. Fill in the blanks (worst case, one guess per second):

DICE IN THE SECRET	HOW MANY POSSIBILITIES	TIME AT 1 GUESS PER SECOND
1	6	6 seconds at most
2	36	_____
3	216	_____
4	1,296	about 21 minutes at most
8	1,679,616	about 19 days at most
12	2,176,782,336	about 69 years at most

A computer can guess 10 billion times a second. So we need secrets that are **very** long and **random**.

Build a dice passphrase

ROLL TWO DICE FOR each word (row number, then column number) and write the words:

Word 1: _____ Word 2: _____ Word 3: _____ Word 4: _____

Sentence to help me remember (a silly picture in my head):

True or False (circle one)

1. Changing "a" to "@" makes a password much stronger. **T / F**
2. A famous quote is a great password. **T / F**
3. Using the same password on two sites is risky. **T / F**
4. A password manager remembers long random passwords for me. **T / F**

Ask a grown-up at home

"Do you use a password manager? Can we set up two-step verification on our email?"

If something goes wrong

IF SOMEONE LEARNS YOUR password or uses your account, it is **not your fault**.

Pause. Change it. Tell a trusted adult.

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 1 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

True or False: 1 F, 2 F, 3 T, 4 T.

Table blanks: 2 dice = 36 possibilities (36 seconds at most); 3 dice = 216 (216 seconds, about 3.6 minutes, at most).

Why these answers (say it in plain words): swapping letters for symbols is the first trick guessing tools try; quotes are in attacker word lists; reuse lets one leak open many accounts; a manager remembers long random passwords so you do not have to.

Checked numbers (recomputed): $6^{10} = 60,466,176$; $36^4 = 1,679,616$; $7,776^4 = 3.66 \times 10^{15}$; at 10 billion guesses per second, 5 random words take about 45 years on average and 6 words about 350,000 years. Handout times are worst case; the teacher worksheet times are averages (half the space).